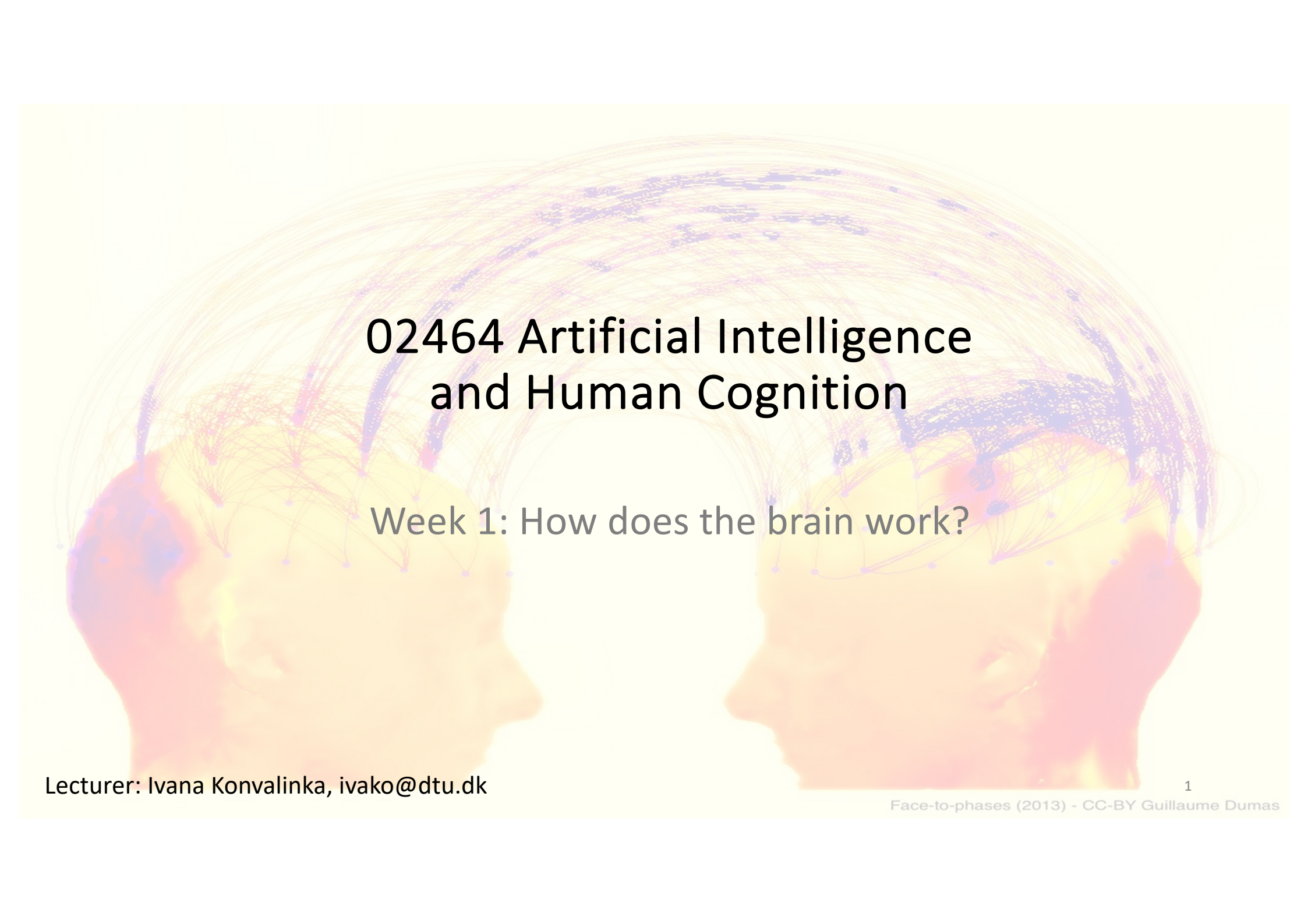




02464 Artificial Intelligence and Human Cognition

Week 1: How does the brain work?

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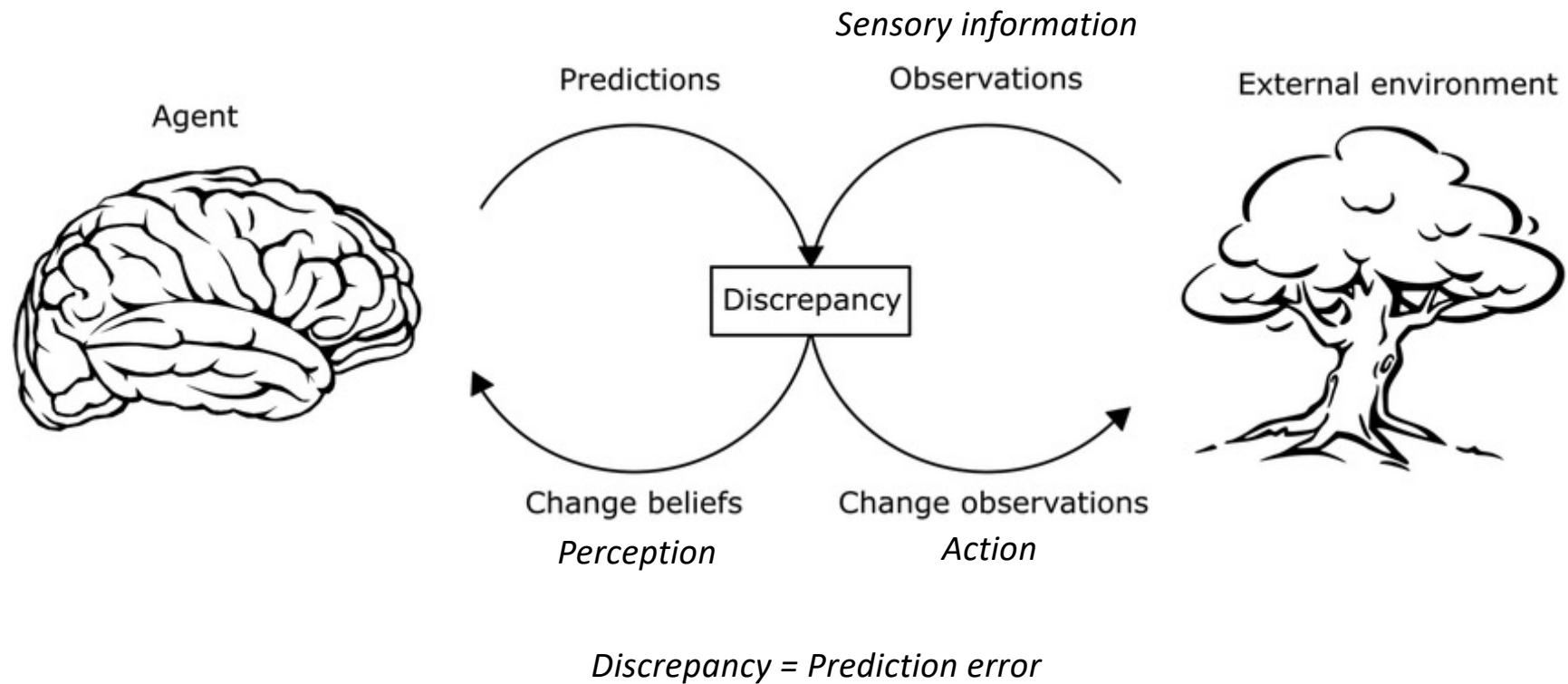
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Face-to-phases (2013) - CC-BY Guillaume Dumas

The brain as a prediction machine

- The brain constantly generates models of the world, by making predictions
- It predicts future actions & sensory input
- It learns internal models of the world to minimize surprise: by making predictions and taking actions to make those predictions come true
- Perception and action work together to minimize uncertainty and surprise:
 - Our predictions – sensory information (perception) = prediction error
- We use prediction errors to guide our beliefs

Active inference – brains minimize prediction error (or surprise)



Active inference – with math

Niche construction optimizes the niche's bound on surprise

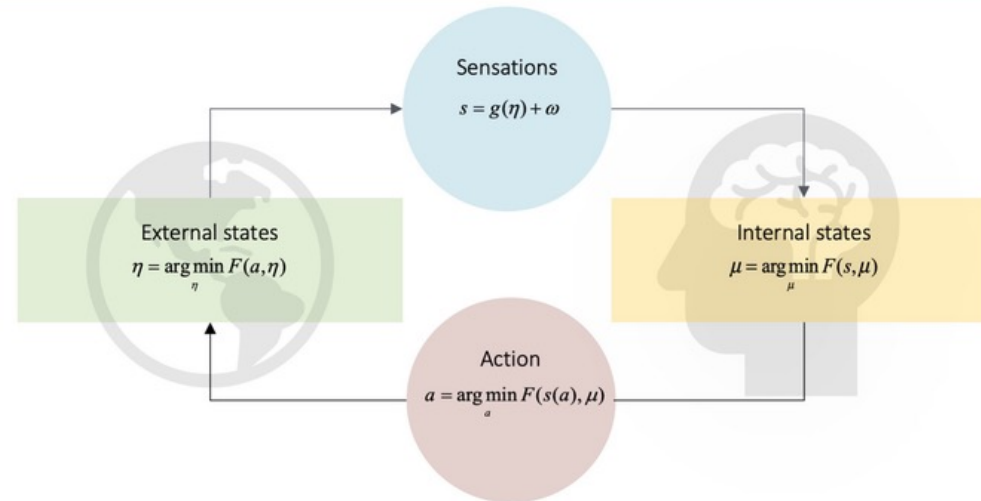
$$F(a, \eta) = \underbrace{D[q_\eta(\mu) \parallel p(\mu | a)]}_{\text{KL divergence}} - \underbrace{\ln p(a)}_{\text{Surprise}}$$

$$\eta = \arg \min_{\eta} \text{Divergence}$$

Sensations minimize the agent's bound on surprise

$$F(a, \eta) = \underbrace{D[q_\eta(\mu) \parallel p(\mu)]}_{\text{Complexity}} - \underbrace{E_q[\ln p(a(s) | \mu)]}_{\text{Accuracy}}$$

$$s = \arg \max_s \text{Accuracy}$$



Action minimize the agent's bound on surprise

$$F(s, \mu) = \underbrace{D[q_\mu(\eta) \parallel p(\eta)]}_{\text{Complexity}} - \underbrace{E_q[\ln p(s(a) | \eta)]}_{\text{Accuracy}}$$

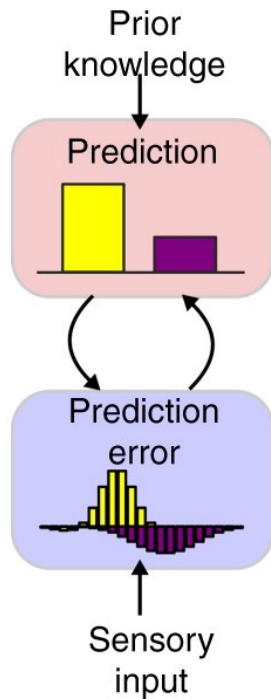
$$a = \arg \max_a \text{Accuracy}$$

Perception optimize the agent's bound on surprise

$$F(s, \mu) = \underbrace{D[q_\mu(\eta) \parallel p(\eta | s)]}_{\text{KL divergence}} - \underbrace{\ln p(s)}_{\text{Surprise}}$$

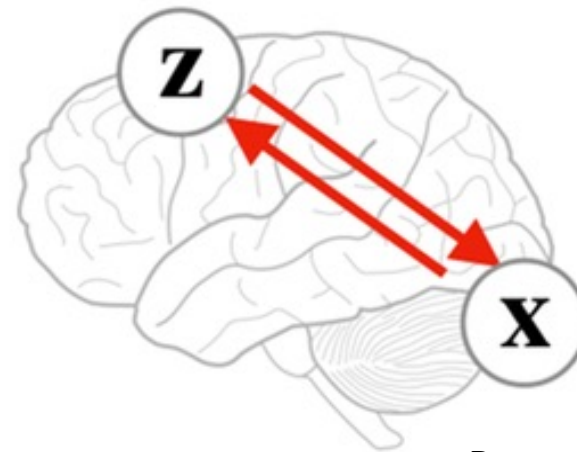
$$\mu = \arg \min_{\mu} \text{Divergence}$$

Top-down vs. bottom-up processing



Top-down processing

Using models, prior experiences, expectations to predict and interpret sensory information



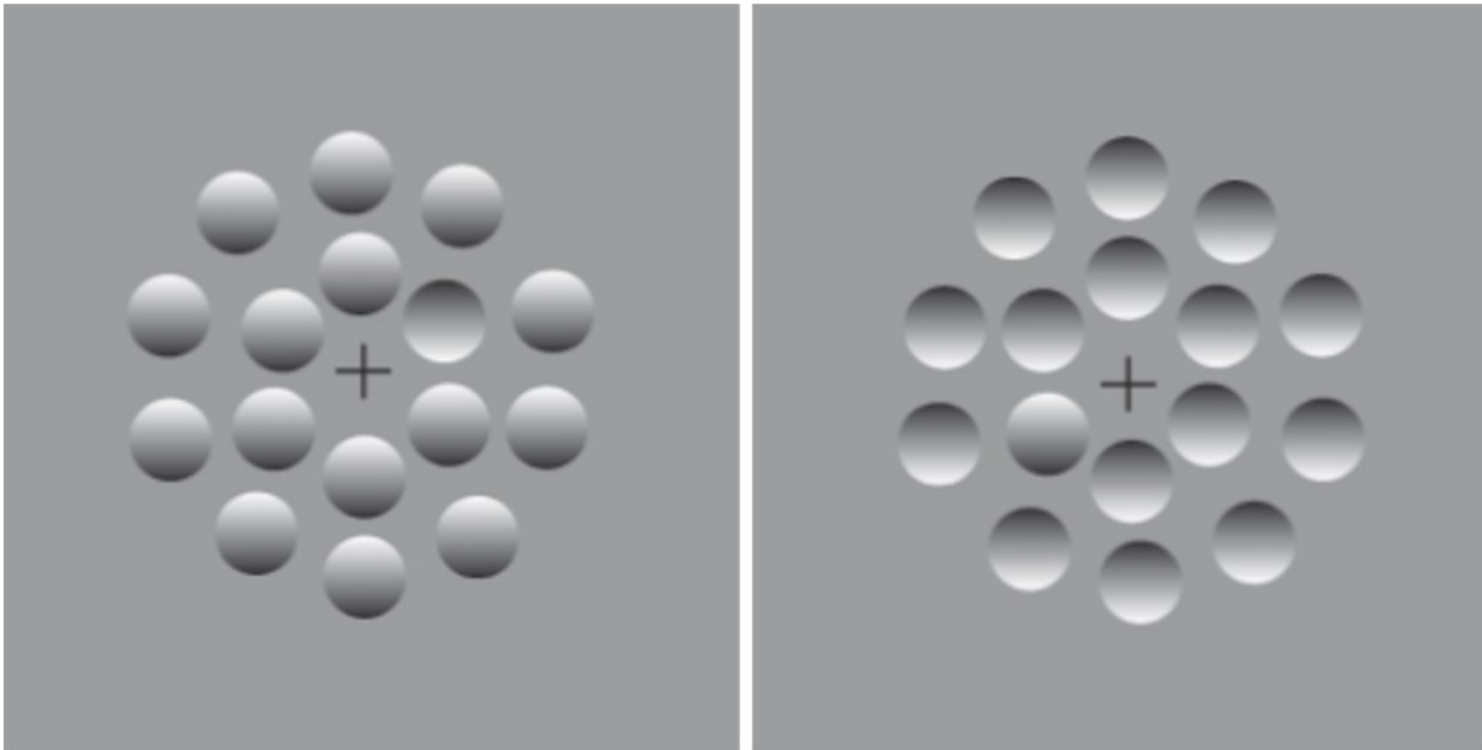
Bottom-up processing

Taking raw sensory input, and building up understanding from sensory details, which are sent back to higher-level brain areas

Top-down processing reading example

i cdnuolt blveiee taht I cluod aulacly uesdnatnrd waht I
was rdanieg. The phaonmneal pweor of the hmuan
mnid, aoccdrnig to a rscheearch at Cmabrigde
Uinervtisy, it dseno't mtaetr in waht oerdr the ltteres in
a wrod are, the olny iproamtnt tihng is taht the frsit and
lsat ltteer be in the rghi t pclae. The rset can be a taotl
mses and you can sitll raed it whotuit a pboerlm. Tihs is
bcuseae the huamn mnid deos not raed ervey lteter by
istlef, but the wrod as a wlohe. Azanmig huh? yaeh and
I awlyas tghuhot slpeling was ipmorantt!

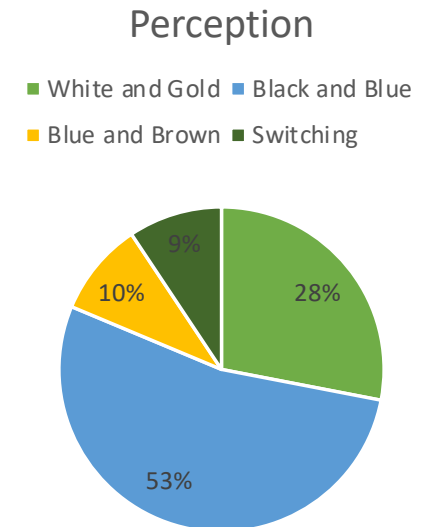
Our (top-down) predictions (priors) guide
how we perceive the world



Our predictions (priors) guide how we perceive the world



Is the dress white and gold...
... or black and blue?



People have different priors about illumination

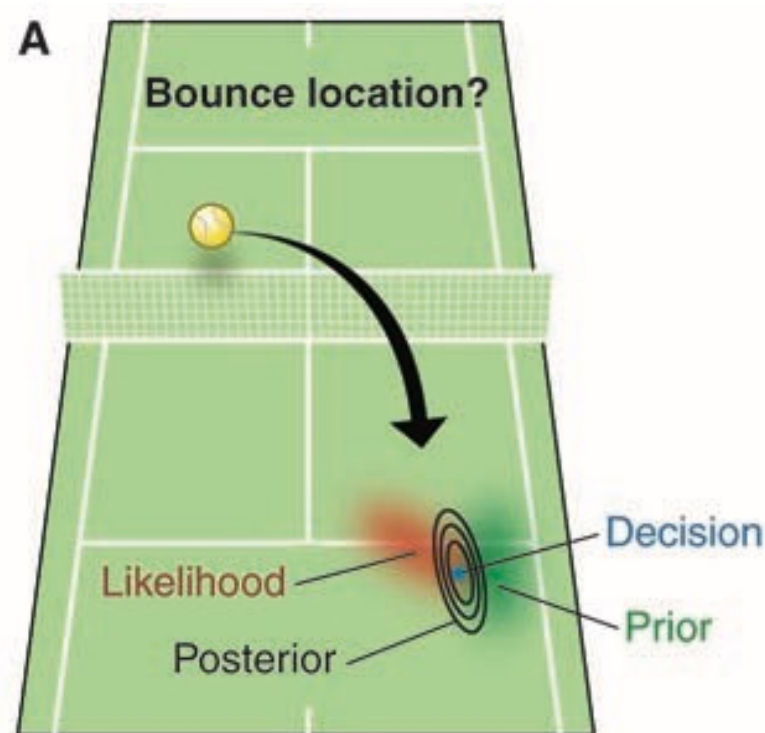
Manipulating these priors before seeing the
photo can steer how observers see the dress

The brain integrates priors (expectations) & sensory input in an optimal way

Bayes Rule:

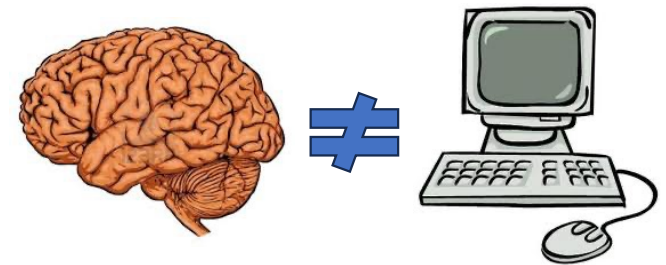
$$\overbrace{P(\text{state}|\text{sensory input})}^{\text{Posterior}} = \frac{\overbrace{P(\text{sensory input}|\text{state})}^{\text{Likelihood}} \overbrace{P(\text{state})}^{\text{Prior}}}{P(\text{sensory input})}$$

If it's raining and your vision is compromised, you will rely more on your prior



The (Bayesian) brain is embodied

- The brain does not work in isolation – it is not sitting there computing probabilities: it has a body
- Cognitive processes and their context unfold continuously and simultaneously in real-time, and in continuous interaction with the environment
- Embodied cognition: the body (and its interactions with the environment) is deeply integrated into cognition and has a central role in shaping the mind: how we think, learn, perceive, speak, interact...
- If we move differently, we end up thinking differently



AI fails when there is a need for embodied cognition

Task:
Choose which piece to
move where

high-level task



**Problems that are difficult
to solve for AI are
associated with the
interface to the real world**

*“The body, in a very simple
way, is the foundation for
intelligent and cautious
action”*

Task:
Physically manipulate
each piece and move it
around the board

low-level task



– Joshua Bongard, NYT
(April 11th, 2023)
Professor of Computer
Science

AI fails when understanding requires understanding of embodied processes

ChatGPT 5 ▾

Could you help me solve a problem? Annie was playing ball yesterday at 10am. Then she disappeared, and her whereabouts were unknown. She was playing ball again at 6pm. Was she alive at noon? (Remember, you have no information about her at this time)

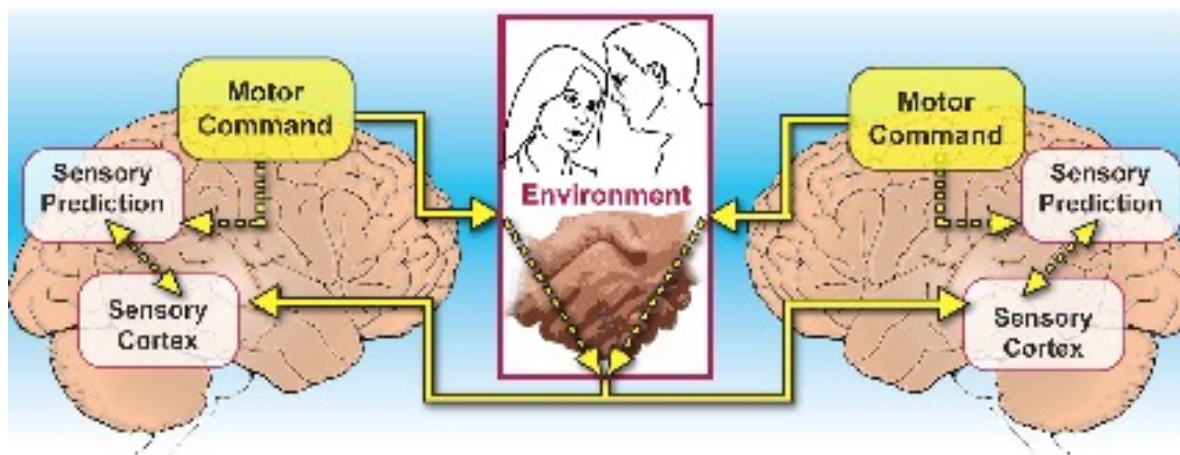
Thought for 11s

Short answer: you can't tell.

Knowing she was alive at 10:00 and again at 18:00 doesn't imply anything about her exact state at 12:00. She could have been alive then—or (counterexample) had a cardiac arrest at 11:50, been clinically dead at 12:00, and resuscitated at 12:30, then played ball at 18:00. So her status at noon is indeterminate from the given info.

The (Bayesian) brain is social

- The brain is not just predicting objects in the environment: it is predicting other people
- The brain predicts another person's actions (content, timing), and their intentions



The brain relies on the exchange of verbal and non-verbal signals to predict actions & intentions

Verbal communication



Non-verbal communication



speech
backchanneling
vocalizations

facial movements & expressions
hand gestures
gaze direction

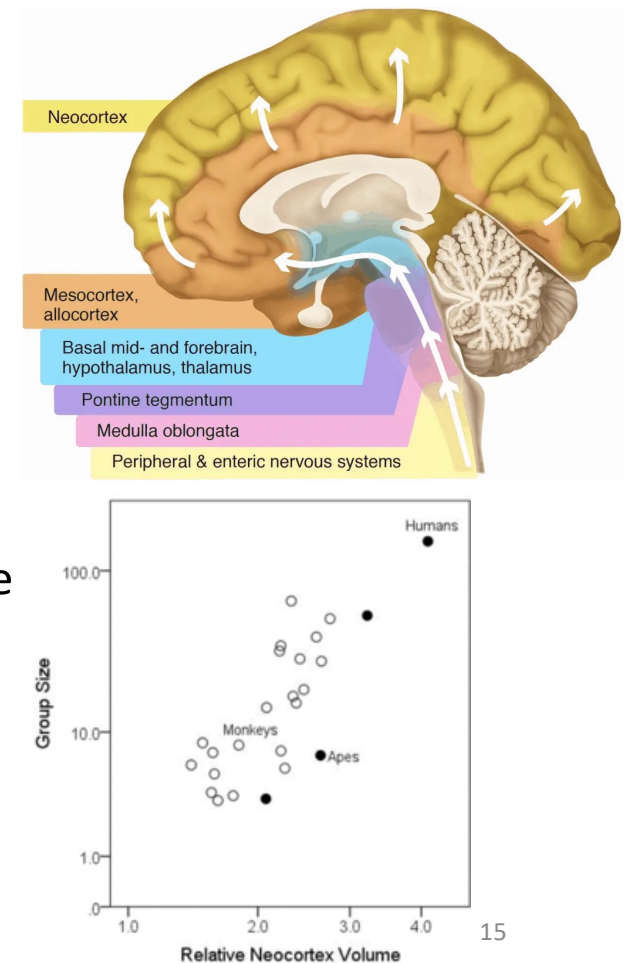
posture
eye movements

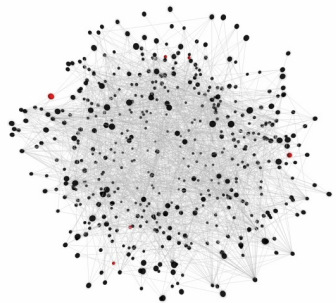
bodily movements

pupil dilation
personal distance
heart rates

The brain is hardwired to interact with other people

- Human brain (neocortex) proportionally much larger in humans compared to other primates/mammals
- Neocortex
 - Outermost layer, largest part of cerebral cortex
 - comprises many brain areas involved in higher social cognition
 - language, empathy, emotion regulation, thinking about others
- Correlation between social group size among primates and relative neocortex volume: *social brain hypothesis*





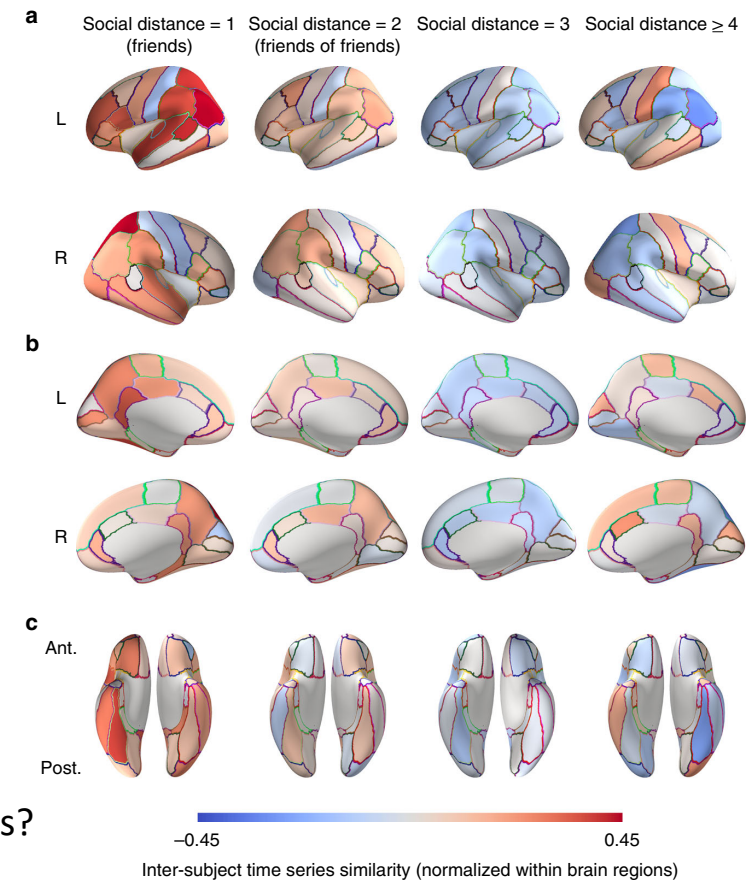
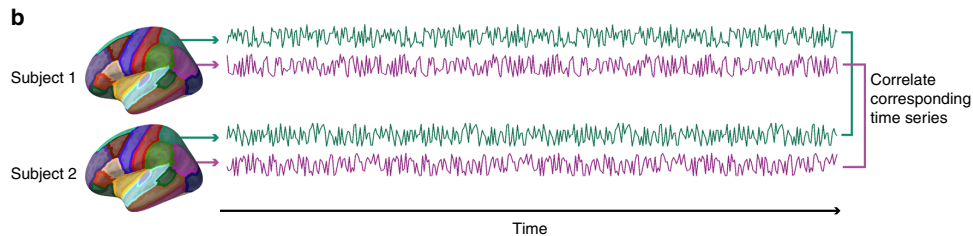
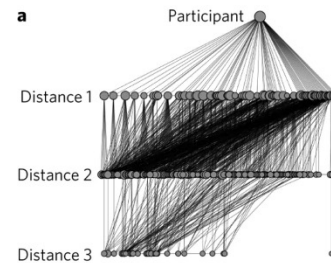
Social networks and the brain

- Social networks shape structural brain properties
 - people with larger social networks have structurally larger social brain areas, and higher white matter connectivity in brain networks
 - causal relationship: living in larger groups causes increases in structural brain areas and functional connectivity between brain areas in monkeys
- Your brain knows your friends' friends: people neurally encode the network position of their peers: eigenvector centrality & social distance

Social networks and the Bayesian brain?

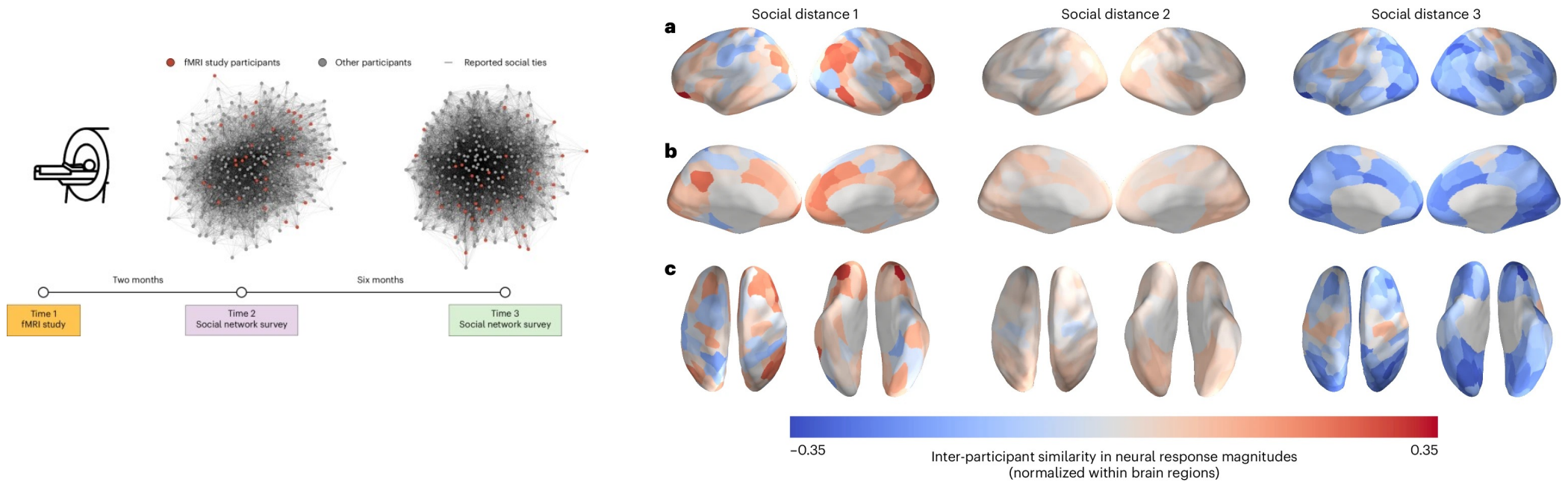
- Your prediction of others is influenced by top-down beliefs and expectations
- If you encode the network position of others, you have expectations of how they will act/interact, what their goals are, etc.
- But you also receive real-time sensory feedback of others actions, which update your predictions (and own actions)

Neural homophily: friends have more similar brain responses



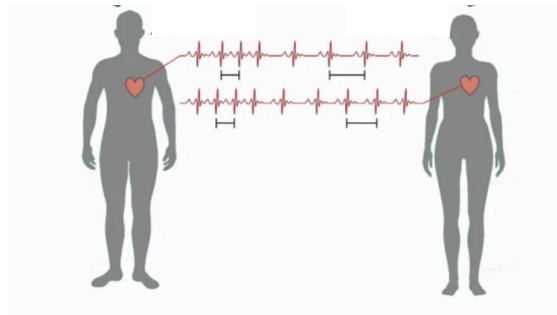
- Do people who become friends have more similar brain responses?
- Does that make them better able to predict each other?

Neural similarity predicts whether strangers become friends



Beyond the brain... social bonds are embodied

- People synchronize bodily rhythms with each other: bodily movements, gaze patterns, postures, heart rates, etc.
- Socially close pairs synchronize more, e.g., mother-baby, romantic couples
- Synchrony predicts formation of bonds
- When people synchronize, they become more cooperative towards each other, like each other better, feel more group cohesion, etc.



Our experiment today

- Do our class social networks influence real-time interaction dynamics, and the coordination between people's bodily signals?
- How much of this is influenced by top-down expectations vs. bottom-up sensory feedback?

